

Superstore BI & Data Analytics

1. Executive Summary

This project aims to build an end-to-end Business Intelligence (BI) solution for a Superstore, starting from raw data engineering and database design, moving through in-depth SQL analysis, and culminating in an interactive Power BI dashboard. The project focuses on extracting actionable insights to optimize profitability and understand customer purchasing behavior.

Key Performance Indicators (KPIs):

- **Total Sales: \$2.30M**
- **Total Profit: \$286.40K**
- **Profit Margin: 12.47%**
- **Total Customers: 793**

2. About the Dataset

Overview:

The analysis is based on the well-known **US Superstore Dataset**, which records comprehensive transactional data for a retail company operating across the United States. It contains 9,994 records representing individual product purchases made over a four-year period.

Dataset Structure & Key Features:

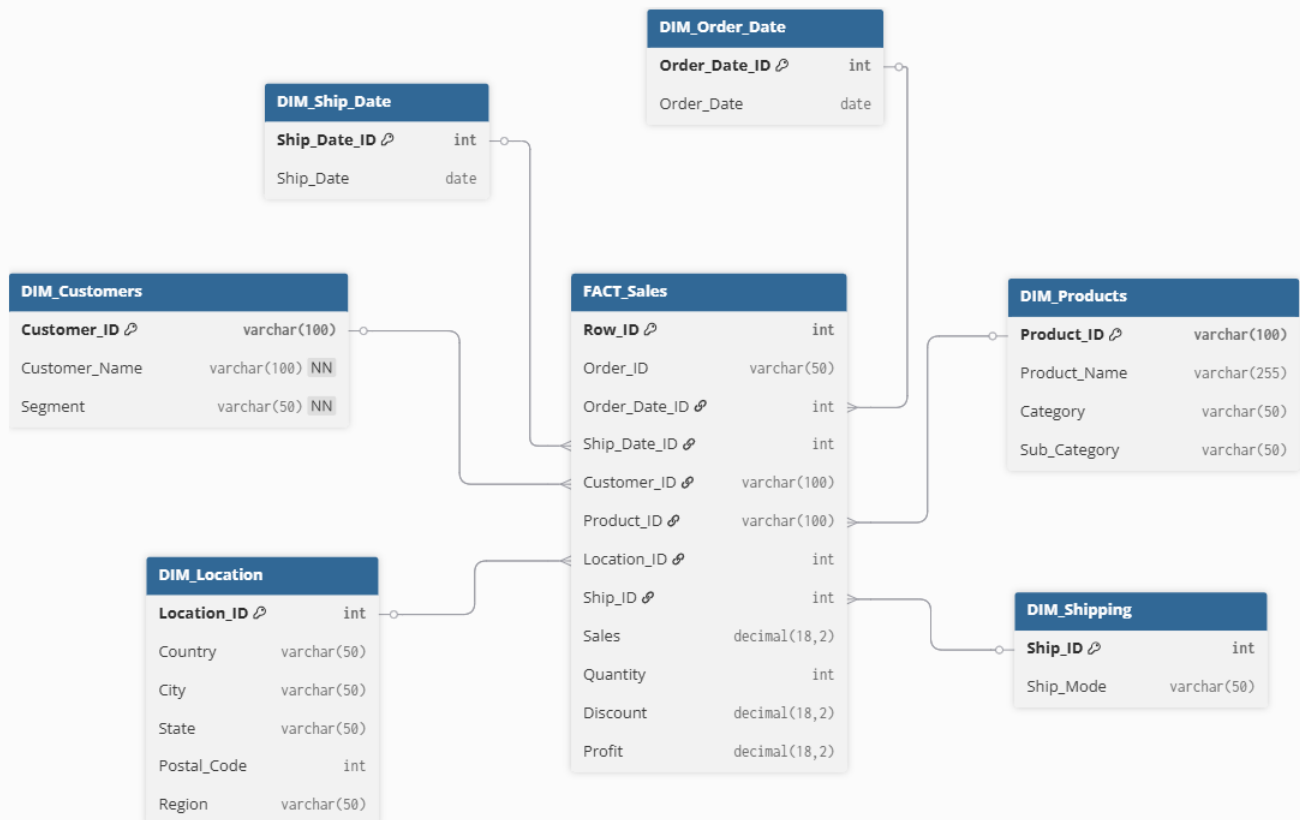
The dataset provides a granular view of the business across multiple dimensions:

- **Financial & Fact Metrics:** Sales (Revenue), Profit (Net income per item), Discount (Percentage reduction in price), and Quantity (Number of items purchased).
- **Customer Details:** Customer ID, Customer Name, and Segment (Consumer, Corporate, Home Office).
- **Product Information:** Product ID, Product Name, Category (Furniture, Office Supplies, Technology), and Sub-Category.
- **Geographic Data:** Country, Region (East, West, Central, South), State, City, and Postal Code.
- **Logistics & Order Details:** Order ID, Order Date, Ship Date, and Ship Mode.

3. Data Engineering & SQL Architecture

The project goes beyond standard data visualization by establishing a robust data infrastructure to ensure analytical accuracy and speed:

- **Data Modeling:** Raw flat-file data was transformed into a professional **Star Schema** within SQL Server. The architecture consists of a central fact table (FACT_Sales) surrounded by 6-dimension tables (Customers, Products, Location, Shipping, Order Date, Ship Date). This design guarantees optimal query performance.



- **Advanced SQL Analytics:** Complex SQL queries were authored to extract deep business rules, including:
 1. **Market Basket Analysis:** Utilized Self-Joins and Common Table Expressions (CTEs) to identify product pairs that are frequently purchased together.
 2. **Yearly Top Products:** Implemented Window Functions (e.g., ROW_NUMBER ()) to dynamically rank the top-selling products for each fiscal year.
 3. **Profitability Audits:** Isolated high-volume, loss-making customers (orders > 5, profit < 0) to highlight structural pricing or discount issues.

4. Power BI & DAX Implementation

Data Analysis Expressions (DAX) were utilized to create custom analytical measures rather than relying on implicit calculations, allowing for highly flexible reporting:

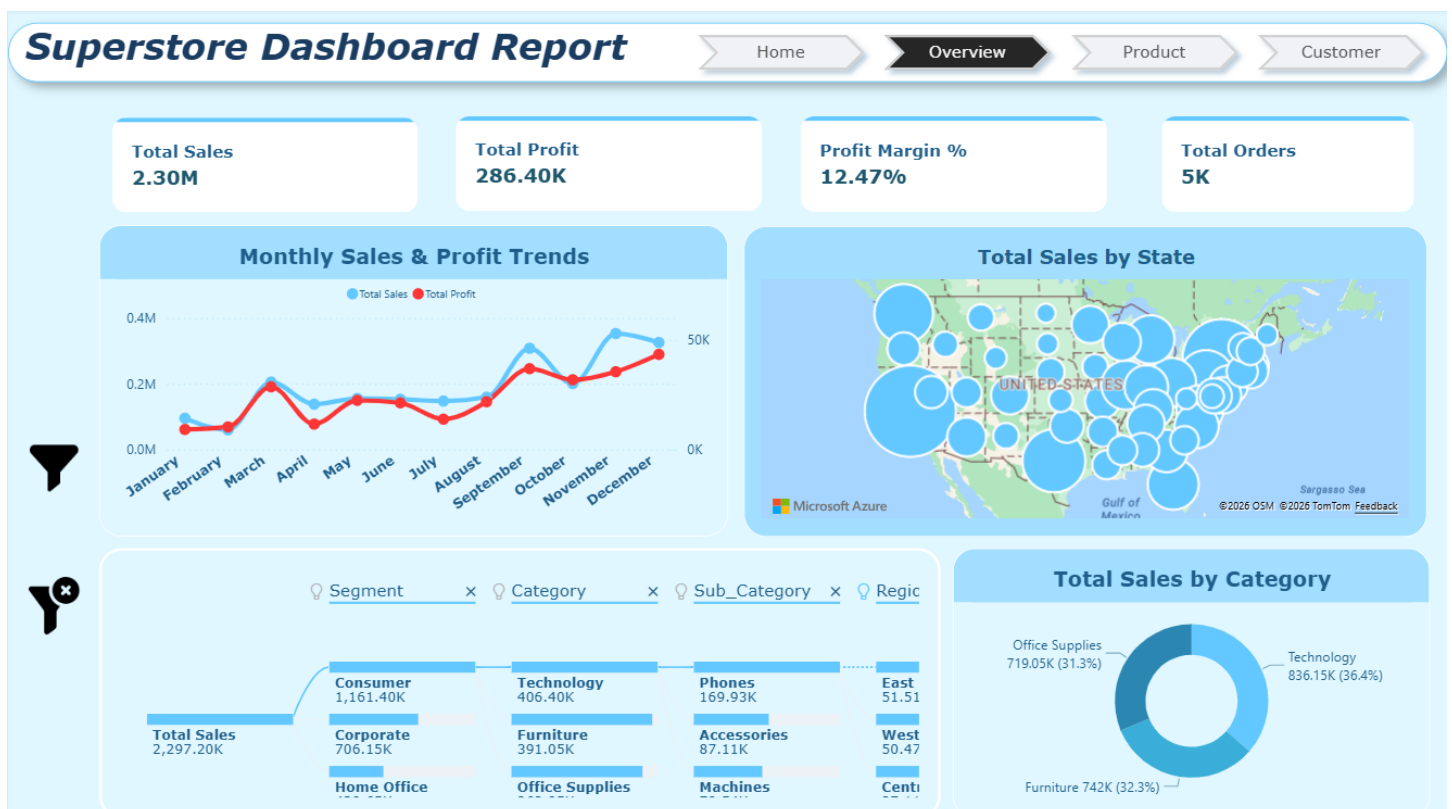
- **Performance Metrics:** Calculated metrics such as Sales Per Customer (\$2.90K).
- **Profitability Metrics:** Calculated overall profit margins and flagged Loss Making Orders to monitor financial health.

5. Dashboard Strategy & Visuals

The reporting layer is divided into interactive pages, each focusing on a distinct business aspect:

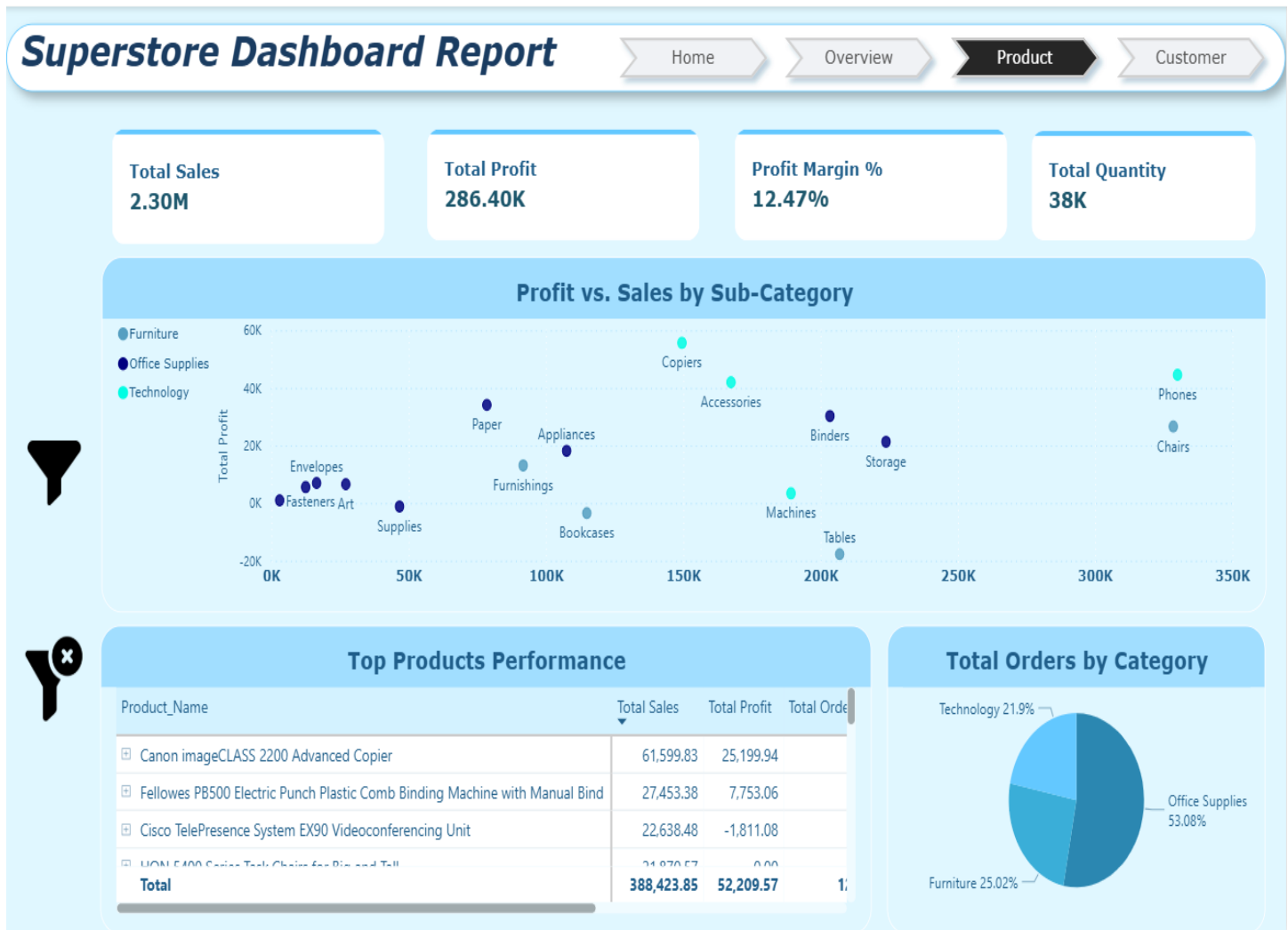
A. Business Overview Page:

- Provides a high-level summary using an Area Chart to monitor seasonal trends and an Azure Map for geographic sales distribution.
- Features an interactive Decomposition Tree to break down sales dynamically from overall segments down to specific sub-categories.



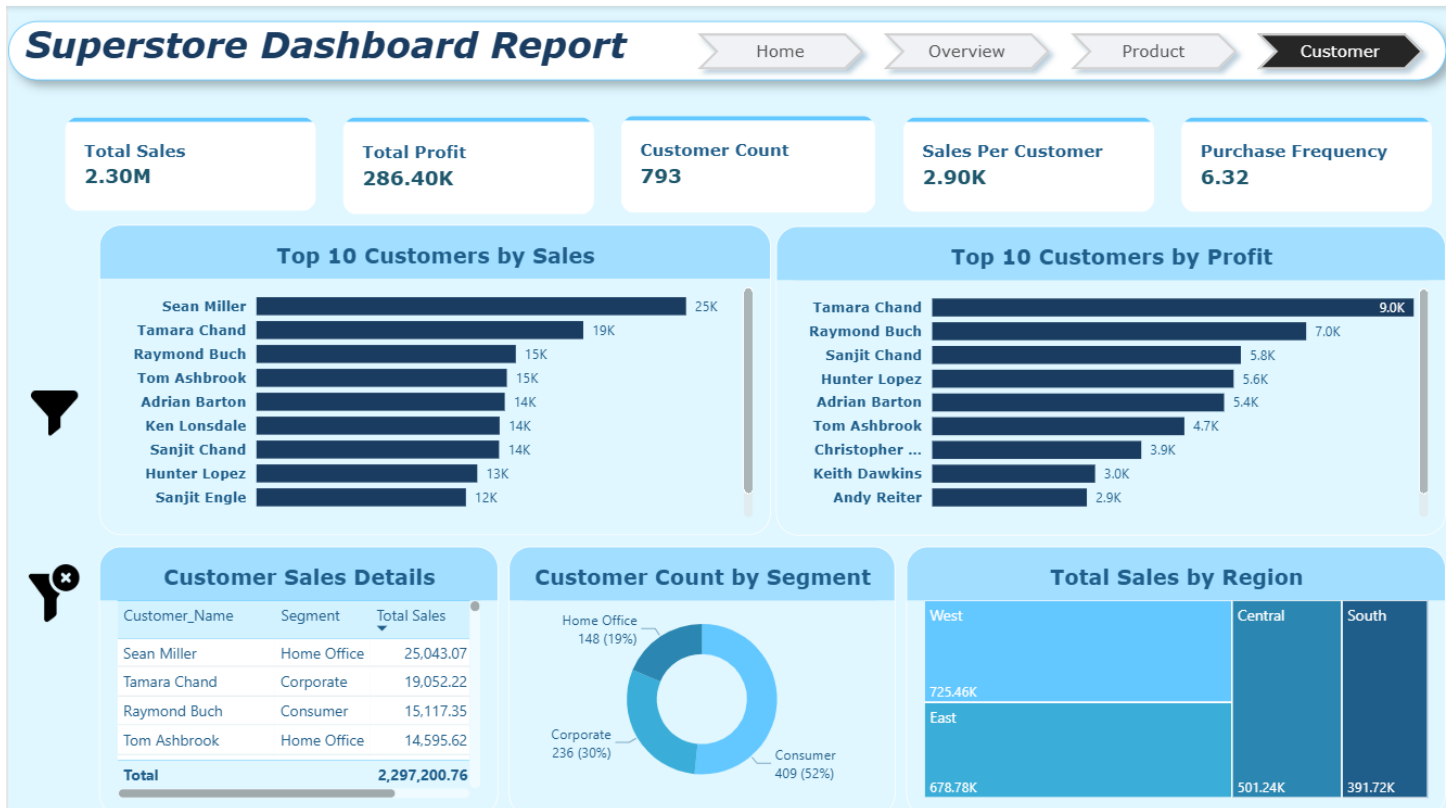
B. Product Performance Page:

- Focus on the performance of 38K sold items.
- Utilizes a Matrix table with conditional formatting (Data Bars) for rapid comparison between gross sales and net profit, easily identifying "Hero" products versus "Bleeders".



C. Customer Insights Page:

- Highlights the top 10 VIP customers by both sales and profit using Dual Horizontal Bar Charts.
- Employs a Donut Chart to illustrate customer segment distribution clearly.



6. Key Business Insights & Recommendations

- Volume vs. Value Paradox:** While the *Office Supplies* category generates the highest order volume (53.08%), the *Technology* sector yields the highest overall revenue share (36.4%).
 - **Recommendation:** Reallocate a larger portion of the marketing budget toward promoting high-margin Technology product bundles to maximize profitability.
- Consumer Segment Dominance:** The *Consumer* segment is the absolute core of the business, representing 52% of the client base (409 out of 793 customers) and capturing the lion's share of total revenue.
 - **Recommendation:** Implement targeted loyalty programs for this specific segment to improve retention rates and increase the average order value per customer.
- Regional Strength:** The *West Region* is the undisputed leader in geographic sales volume.
 - **Recommendation:** Investigate the specific marketing strategies and operational factors driving success in the West and replicate these frameworks in underperforming regions.